10.2 Homework

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Question 1

0/1 pt

Find c if $a = 2.72$ mi, $b = 3.31$ mi and $\angle C = 40.4$ degrees.
Enter c rounded to 3 decimal places.

$c =$ 2.15 mi;

Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question Help: Video

Question 2

1/1 pt 4

Find c if $a = 2.7$ mi, $b = 3.42$ mi and $\angle C = 40.4$ degrees.

$c =$ 2.21863 mi;

Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question Help: Video

Question 3

1/1 pt 4

Solve the triangle if $a = 28$ yd, $b = 53$ yd and $c = 30$ yd.

$\alpha =$ 23.101 °

$\beta =$ 132.04 °

$\gamma =$ 24.859 °

Assume $\angle\alpha$ is opposite side a, $\angle\beta$ is opposite side b, and $\angle\gamma$ is opposite side c.
Enter your answer as a number; answer should be accurate to 2 decimal places.

Question 4

1/1 pt 4

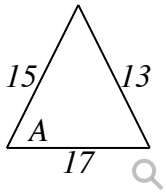
Find a if $b = 163$ km, $c = 149$ km and $\angle\alpha = 85^\circ$.

$a =$ 211.03672 km;

Assume $\angle\alpha$ is opposite side a, $\angle\beta$ is opposite side b, and $\angle\gamma$ is opposite side c.

Question 5

1/1 pt 2

Given the triangle 

, find the measure of angle A using the Law of Cosines. *Picture is not drawn to scale*

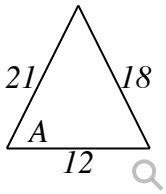
$A =$ 47.43 degrees

Give your answer accurate to at least one decimal place

Question Help: Video

Question 6

1/1 pt 4

Given the triangle 

, find the measure of angle A using the Law of Cosines. *Picture is not drawn to scale*

$A =$ 58.81 degrees

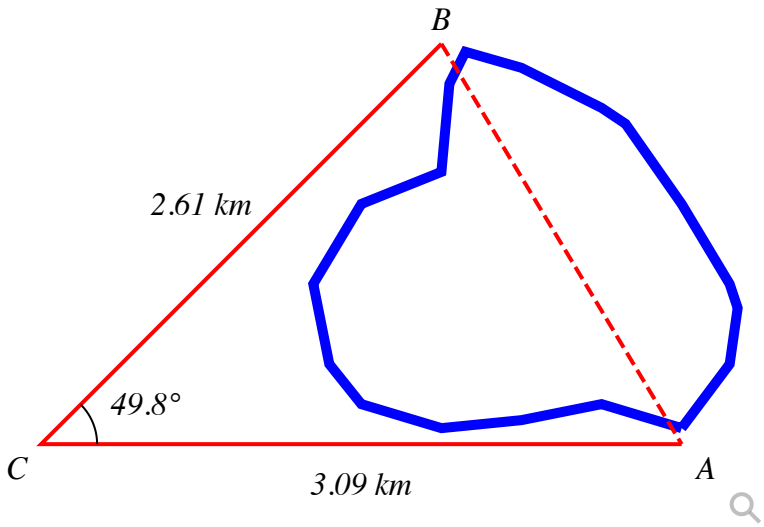
Give your answer accurate to at least one decimal place

Question Help: Video

Question 7

0/1 pt 4

To find the distance across a small lake, a surveyor has taken the measurements shown. Find the distance across the lake using this information.



NOTE: The triangle is NOT drawn to scale.

distance = 307.32 km

Enter your answer as a number; your answer should be accurate to 2 decimal places.

Question Help: Worked Example 1

Question 8

1/1 pt 4

A triangular field has sides of lengths 23, 56, 73 mi.
Enter your answer as a number; answer should be accurate to 2 decimal places.

Find the largest angle in degrees:

130.24 °

Question Help: Worked Example 1

Question 9

1/1 pt 2

A pilot flies in a straight path for 1 h 30 min. She then makes a course correction, heading 10 degrees to the right of her original course, and flies 2 h in the new direction. If she maintains a constant speed of 600 mi/h, how far is she from her starting position?

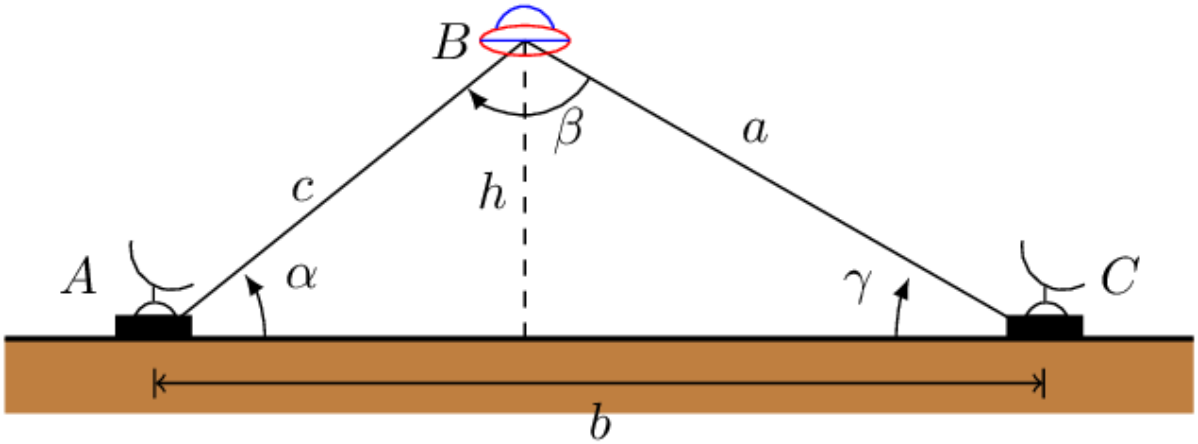
Your answer is 2092.17 mi;

Enter your answer rounded to two decimal places.

Question Help: Worked Example 1

Question 10

1/1 pt 4



NOTE: The picture is NOT drawn to scale.

To find the distance between two radar installations a UFO calculates the distance to installation `A` to be `c = 124` km, the distance to installation `C` to be `a = 47` km, and the angle between them `beta = 31^@` . Find the distance between the installations.

Distance between the installations = 87.14274 km

Enter your answer as a number; your answer should be accurate to 2 decimal places.

Question Help: Video

Question 11

1/1 pt 4

A steep mountain is inclined 74 degree to the horizontal and rises to a height of 3400 ft above the surrounding plain. A cable car is to be installed running to the top of the mountain from a point 910 ft out in the plain from the base of the mountain. Find the shortest length of cable needed.

Your answer is 3887.54 ft

Question Help: Video

Question 12

0/1 pt

A parallelogram has sides of lengths 6 and 4, and one angle is 31°.

What is the length of the smaller diagonal?

length = 3.293 units

What is the length of the longer diagonal?

length = 9.653 units

Enter your answer as a number; your answer should be accurate to 3 decimal places.

Question Help: Worked Example 1